Lesson notes · for you

AI security engineering: unit check

AI security · University · Unit check

Time	Materials
45 min	Paper and a pen. No device needed.

Learning goals

- Threat-model an AI application
- The OWASP Top 10 for LLM applications
- Attacking and hardening models
- Red teaming and responsible disclosure

Activity: Design review

1. Teams of three get one made-up AI app
2. Five minutes: its top three risks and their controls
3. Swap: the other team tries to break your plan, on paper
4. Agree the fixes, and the test that proves each one

Everything stays on paper and made-up: no live systems.

Discussion

- Which practice will you use first on your own work?
- Which question surprised you, and why?
- What would you tell a classmate about testing systems without permission?

Check yourself

- My first practice from this unit, and the project I'll use it on

Try it this week
Pick the practice from this unit you'll use first on something you're building (a threat model, an OWASP review, a robustness test or an evaluation harness) and use it this week.

Worksheet

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Name _____ Class _____ Date _____

1 **In a threat model of an LLM app, why isn't the model a trust boundary?**

- ◯ Anything in its context can steer what it says and which tools it calls
- ◯ It is too slow to check its inputs
- ◯ It runs on someone else's computer

2 **Match each OWASP risk to an example of it.**

LLM03 Supply chain	• • • •	An invented reference ends up in an essay
LLM05 Improper output handling	• • • •	Its instructions, with a password inside, are read out
LLM07 System prompt leakage	• • • •	A reply pasted into a page runs as script
LLM09 Misinformation	• • • •	A look-alike model file with a backdoor

3 **What does this print?**

```
flows = [("model", "email"), ("web", "model")]
reach = {"web"}
for a, b in flows:
    if a in reach:
        reach.add(b)
print("email" in reach)
```

- ◯ True
- ◯ False
- ◯ web

4 **Telling a model "never reveal the password" is enough to keep a password in its system prompt safe.**

- ◯ True
- ◯ False

5 **A linear model has weights 3 and -2. What does this print?**

```
w, x, eps = [3, -2], [2, 1], 0.5
s = sum(a * c for a, c in zip(w, x))
drop = eps * sum(abs(a) for a in w)
print(s, s - drop)
```

- ◯ 4 1.5
- ◯ 4 6.5
- ◯ 4 3.5